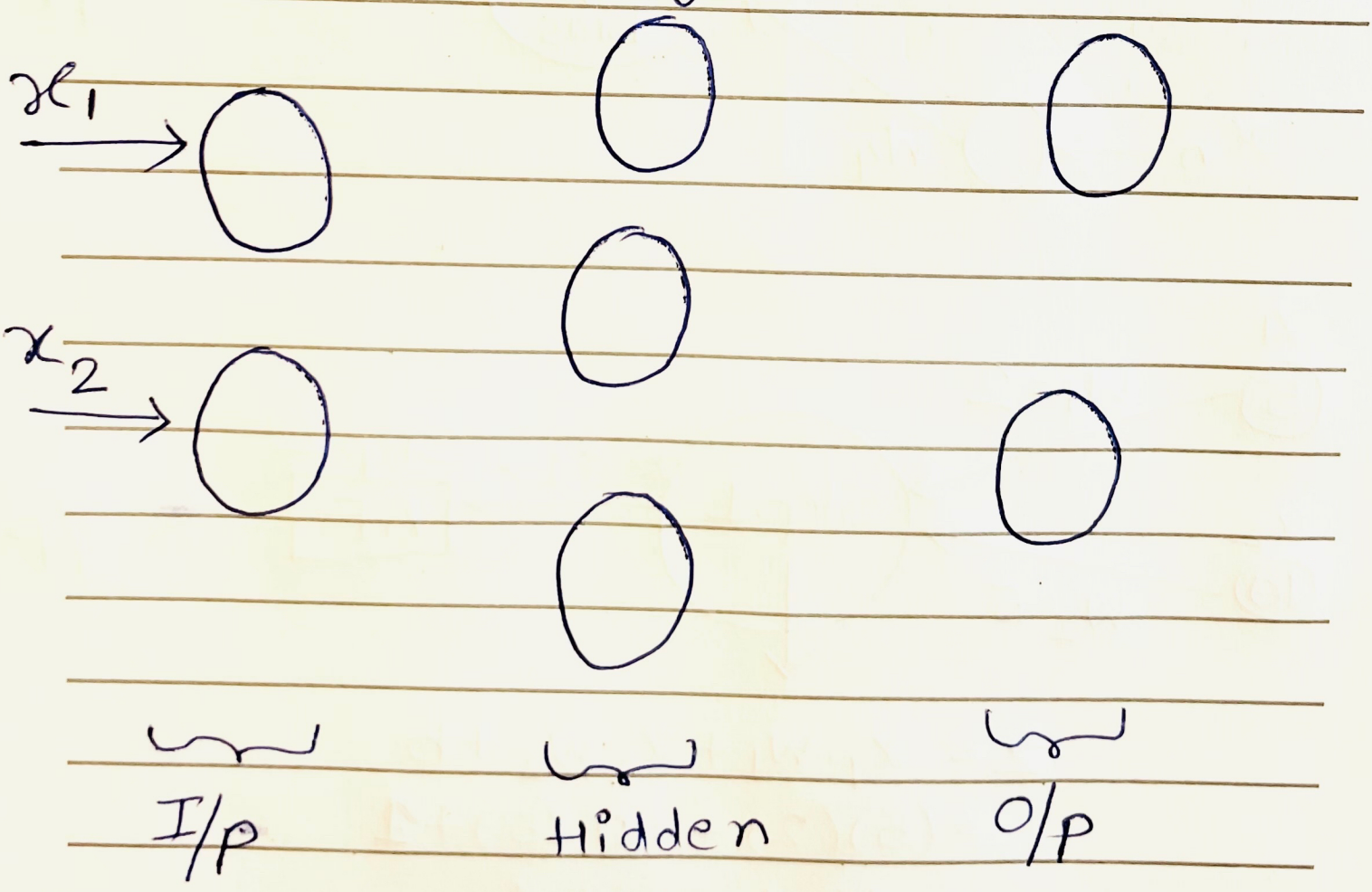
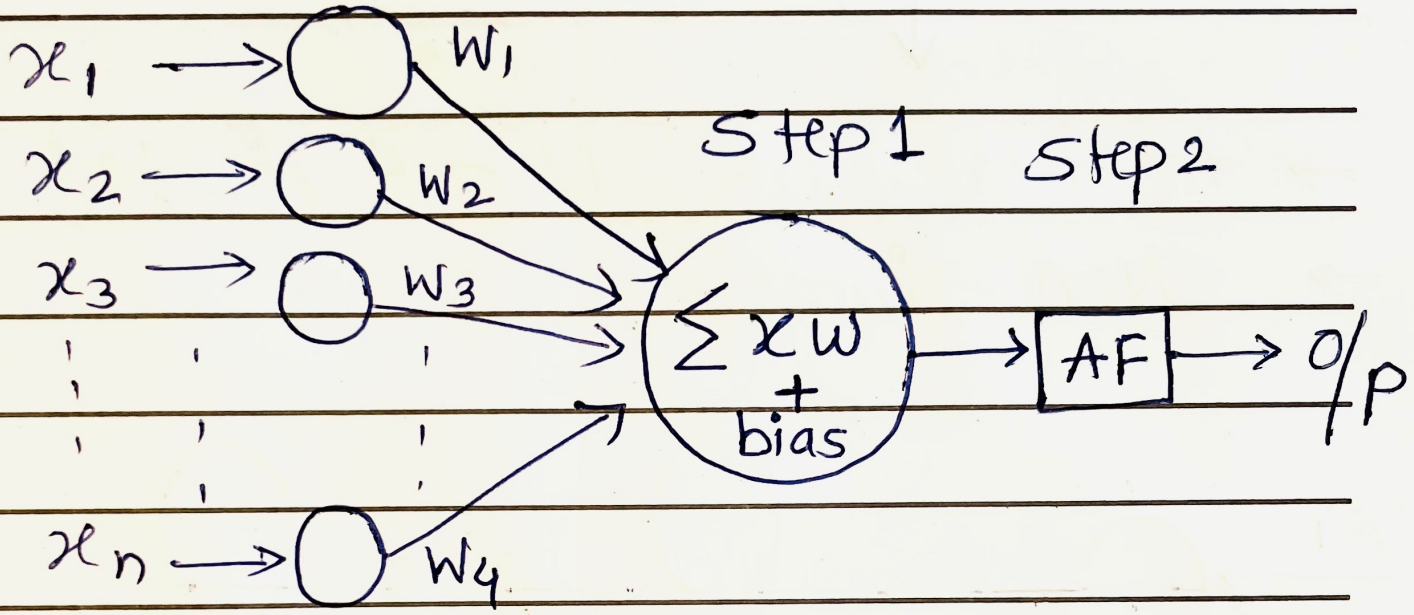
"Activation function"

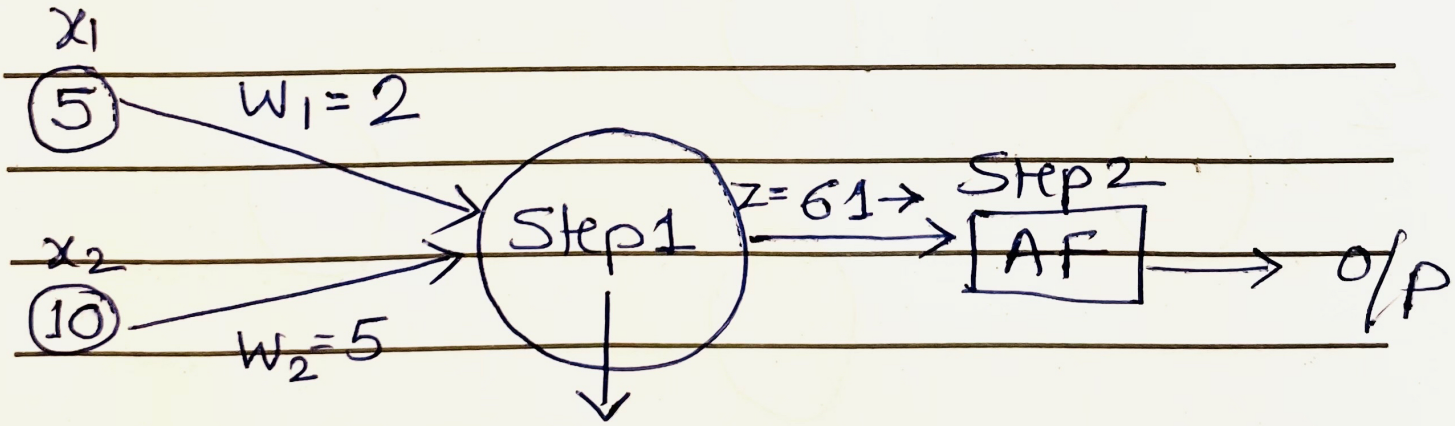
→ Activated OR Not Activated

→ Non-linearity



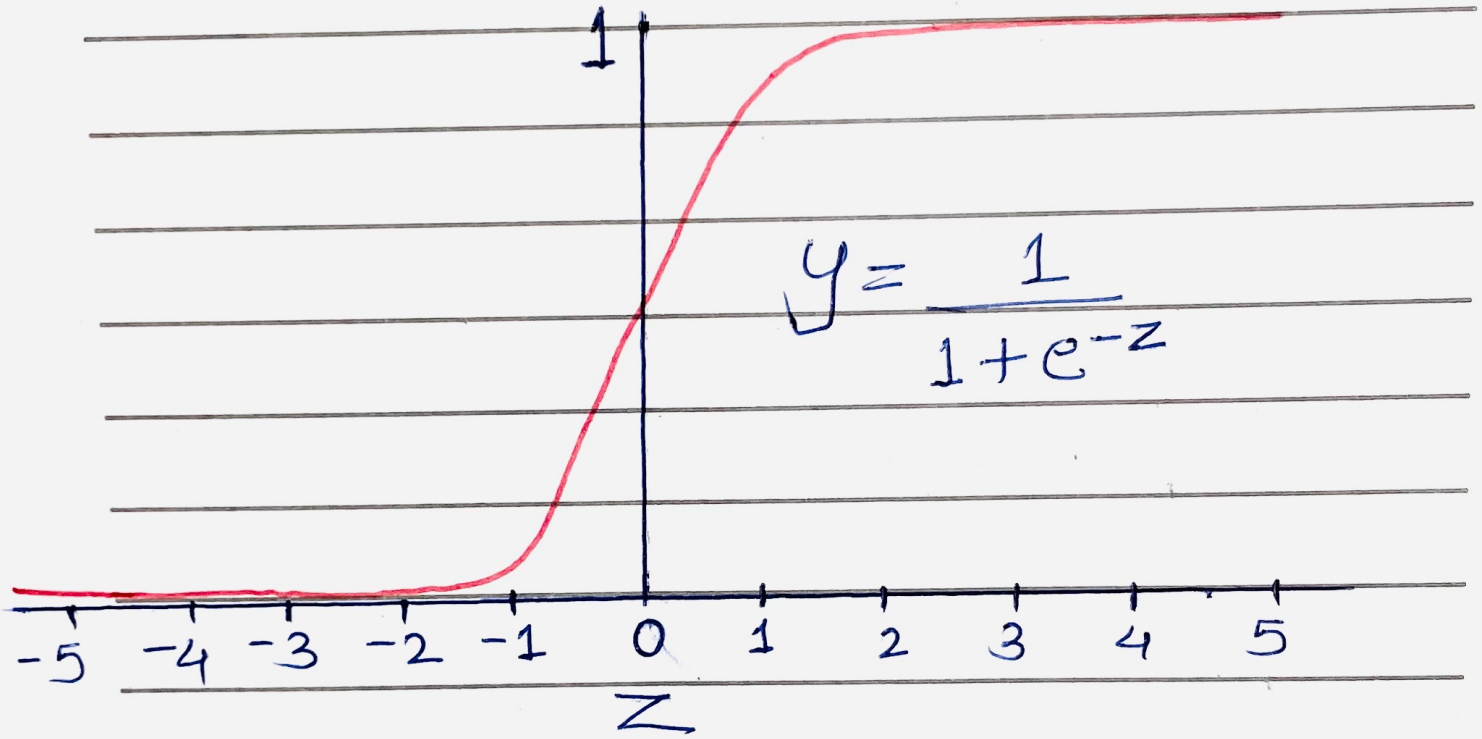
# Input



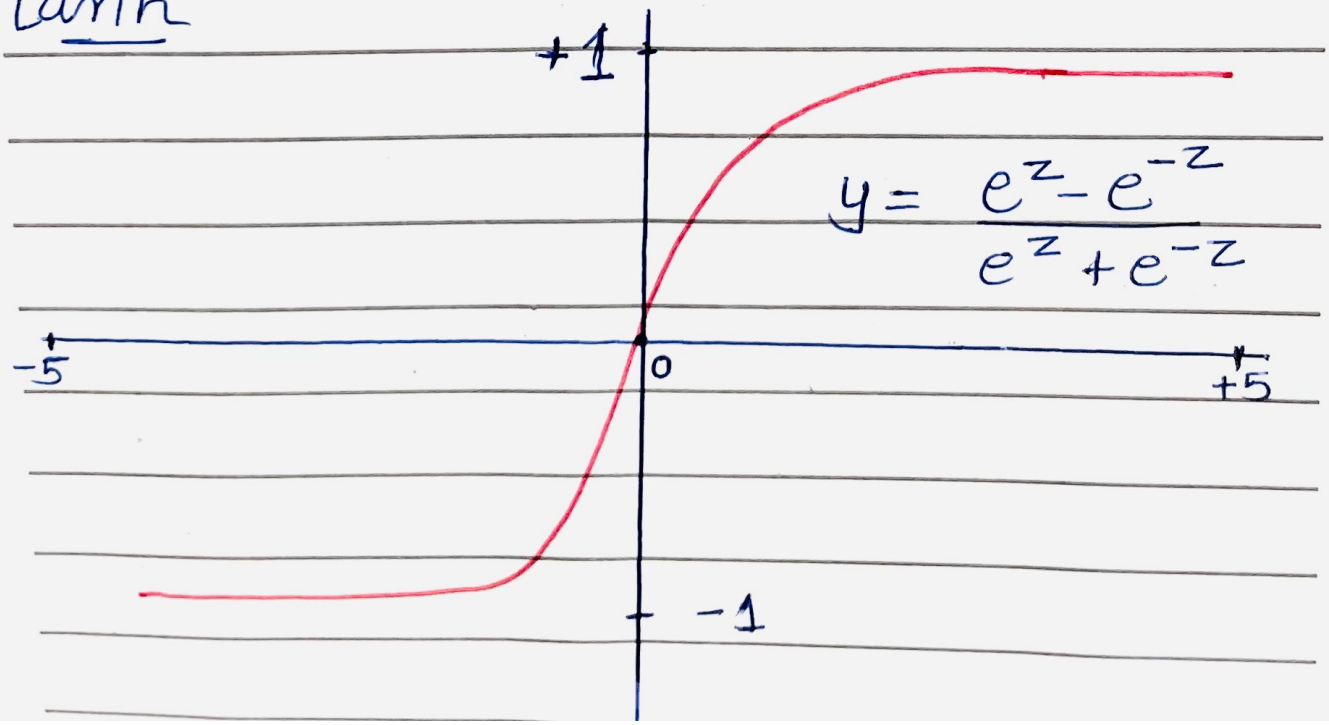




$$z = x_1 \cdot w_1 + x_2 \cdot w_2 + b$$
$$= (5)(2) + (10)(5) + 1$$
$$= 10 + 50 + 1$$
$$= 61$$

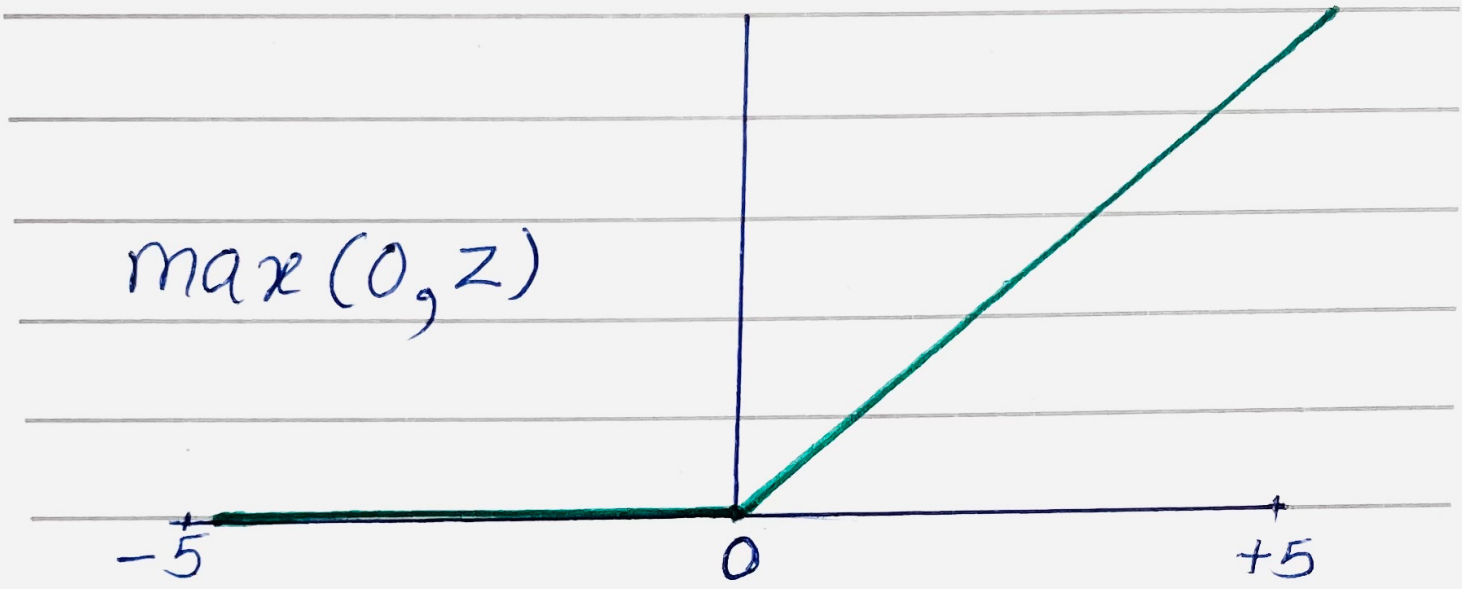
→ Sigmoid



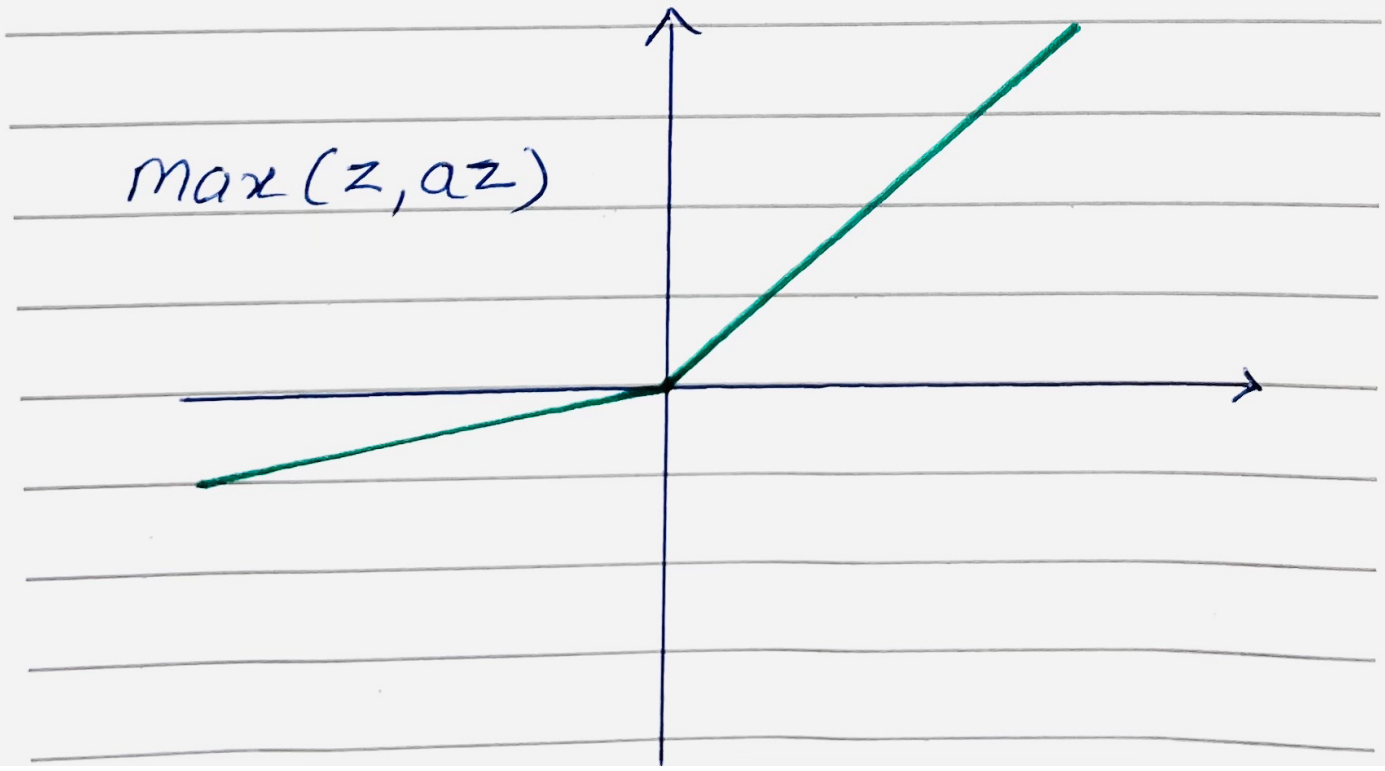

→ tanh



→ ReLU ✓




→ Leaky ReLU



→ Softmax

$$f(z) = \frac{e^{z_i}}{\sum_{j=1}^{k} e^{z_j}}$$

$z = 2 \longrightarrow$ (C1)

$z = 3 \longrightarrow$ (C2)

O/P

$$C1 \rightarrow \frac{e^2}{e^2 + e^3} = \frac{7.38}{7.38 + 20.08} = 0.26$$

$$C2 \rightarrow \frac{e^3}{e^2 + e^3} = \frac{20.08}{7.38 + 20.08} = 0.74$$